

sloggs — MIDI Implementation Chart

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This document describes the MIDI behavior of the **sloggs** MIDI implementation, an emulation of the Microsoft GS Wavetable Synth.

Updates along with **SPEC.adoc**.

1. Receive data

In every byte pattern below, **n** is the MIDI channel nibble (**0–F**, **0**=channel 1 ... **F**=channel 16).

1.1. Channel Voice Message

8n kk vv — Note off

- **kk** = note number (**00–7F**, 0–127)
- **vv** = release velocity (**00–7F**)

Received, **ignored**

9n kk vv — Note on

- **kk** = note number (0–127)
- **vv** = velocity (**01–7F**, 1–127)

Sets loudness on a fixed curve (perceptually, amplitude scales with the square of velocity/127)

An kk vv — Polyphonic Key Pressure (aftertouch)

Not recognized. Received, no audible effect.

Bn cc vv — Control Change

See **Control Change**.

Cn pp — Program Change

- **pp** = program number (**00–7F**, 0–127)

Combines with whatever Bank Select value is currently in effect (**Control Change**) to select an instrument.

Dn vv — Channel Pressure (aftertouch)

Not recognized. Received, no audible effect.

En ll mm — Pitch Bend Change

- **ll** = LSB
- **mm** = MSB
- Combine to a 14-bit value $mm \times 128 + ll$.
- Center is **00 40** (8192, no bend)

- Applied continuously to every sounding note on the channel — a bend arriving mid-note is heard **immediately**, not just on the next note. Range is set by RPN0, default ± 2 semitones (**Control Change**).

1.2. Control Change

Only the controllers below are recognized. **Every other CC number is received and produces no effect, silently — never an error.**

Bn 00 mm — Bank Select MSB

See Bank Select LSB.

Bn 20 ll — Bank Select LSB

- Received and stored, but has **no effect until a GS Reset System Exclusive has been received at least once.**
- Takes effect on **next Program Change**, and *only* after the SysEx.

Bn 01 vv — Modulation Wheel

Recognized.

Bn 06 mm — Data Entry MSB

See Data Entry LSB.

Bn 26 ll — Data Entry LSB

Recognized only together with RPN 0/1/2 selection, below; meaningless on their own.

Bn 07 vv — Channel Volume

Recognized. Default **64** (100).

Bn 0A vv — Pan

Recognized. Default **40** (64, center).

Bn 0B vv — Expression

Recognized. Default **7F** (127).

Bn 40 vv — Hold 1 (Sustain)

Recognized.

- **00–3F** = off
- **40–7F** = on (default)

Bn 5B vv — Effects 1 Depth (Reverb Send)

Not recognized. Received w/o effect.

Bn 5D vv — Effects 3 Depth (Chorus Send)

Not recognized. Received w/o effect.

Bn 62 11 — NRPN LSB

See NRPN MSB.

Bn 63 mm — NRPN MSB

Recognized only as "deselect the current RPN". **No NRPN parameter is ever addressable** — Data Entry sent after an NRPN selection has no effect.

Sending either, with any value, immediately clears whatever RPN was previously selected to Null.

Bn 64 11 — RPN LSB

See RPN MSB.

Bn 65 mm — RPN MSB

Selects a Registered Parameter for the following Data Entry (CC6/CC38) messages to target:

RPN (MSB,LSB)	Parameter	Applied
00 00	Pitch Bend Sensitivity. Data Entry MSB = semitones (0–24); LSB ignored. Default 2 semitones.	Live — affects notes already sounding.
01 00	Channel Fine Tuning, ±100 cents (14-bit Data Entry).	Latched at Note On. Changing it while a note is held does not retune that note.
02 00	Channel Coarse Tuning, ±64 semitones (Data Entry MSB only; LSB ignored).	Latched at Note On, same as Fine Tuning.
7F 7F	Null (no parameter selected).	—
Others	Any	Stored, but no effect

Bn 78 00 — All Sound Off

Recognized. Releases (does not hard-cut) every sounding note on the channel, **bypassing** a held sustain pedal.

Bn 79 00 — Reset All Controllers

Recognized Bank/Program and the RPN/NRPN selection are left untouched, otherwise re-arms:

Parameter	Value
Channel Volume	100
Pan	64 (center)
Expression	127
Pitch Bend	8192 (center)
Modulation	0
Sustain pedal	Released

WARNING

Reset All Controllers (CC121), GS Reset, GM System On/Off and

System Reset (FF) **DO NOT reset** RPN0/1/2 or which RPN is currently selected — the selection and tuning survive every reset path.

Bn 7A 00 — Local Control

Not applicable — **slopps** has no local keyboard to disconnect. Received, no effect.

Bn 7B 00 — All Notes Off

Recognized. Releases every sounding note on the channel, **honoring** a held sustain pedal (notes under a held pedal stay held).

Bn 7C 00 — Omni Mode Off

See Omni Mode On.

Bn 7D 00 — Omni Mode On

Not recognized. Received, no mode change ([Channel Mode Message](#)).

Bn 7E mm — Mono Mode On (Poly Off)

Recognized. Switches the channel to mono and silences every currently-sounding note on it (implicit all-notes-off, per the MIDI spec).

mm (requested mono voice count) is **ignored** — the channel always behaves as if **mm**=1.

Bn 7F 00 — Poly Mode On

Recognized. Switches the channel to poly and silences every currently-sounding note on it (implicit all-notes-off).

1.3. Channel Mode Message

Mode is only ever Omni Off (each channel answers exactly the Part(s) mapped to it — see Basic Channel in the [chart](#), below). Omni On/Off messages are received but never change this. Mono/Poly are switchable per-channel via CC126/CC127 ([Control Change](#)); a channel is otherwise Poly by default.

1.4. System Common Message

Song Position Pointer, Song Select and Tune Request: **not recognized.** **slopps** renders a Standard MIDI File on its own clock and has no use for sequencer-sync messages.

1.5. System Real Time Message

F8 Timing Clock, FA Start, FB Continue, FC Stop

Not recognized.

FE Active Sensing

Not recognized — received, no effect, and no 300 ms watchdog timeout is armed by receiving it (irrelevant, since **slopps** never transmits it either).

FF Reset

Recognized. Restores every power-on default:

Parameter	Value
Bank/Program	0 / 0 / 0
Channel Volume	100
Pan	64 (center)
Expression	127

Parameter	Value
Pitch Bend	8192 (center)
Modulation	0
Per-Part tuning grid	Flat (0)
GS mode	Cleared (Bank Select inert until the next GS Reset)

WARNING

Same exception as Reset All Controllers (CC121, [Control Change](#)): the currently-selected RPN/NRPN parameter and RPN0/1/2's tuned values survive a Reset unchanged.

1.6. System Exclusive Message

Every recognized SysEx message below uses Roland's **DT1** ("Data set 1") envelope, one parameter address per message:

F0 41 dev 42 12 aa bb cc dd...dd sum F7	Roland GS-family message. dev = device ID — not checked aa bb cc = 21-bit parameter address. dd...dd = data bytes for that address. sum = checksum — not checked
F0 7E dev 09 01 F7	GM System On (Universal Non-Realtime). dev not checked.
F0 7E dev 09 02 F7	GM System Off (Universal Non-Realtime). dev not checked.
F0 7F dev 04 01 11 mm F7	Master Volume (Universal Realtime). dev not checked. Only mm (MSB) is used; 11 (LSB) is never read.

Recognized **DT1** addresses (**aa bb cc**):

Address	Parameter	Effect
40 00 7F	GS Reset	Full device reset, and sets GS mode (Bank Select CC0/CC32 start taking effect).
40 1n 02	RCV CHANNEL (GS Mode)	Remaps which incoming channel Part n listens to. Persists until a Reset.
40 1n 15	USE RHYTHM PART (GS Mode)	Nonzero = Part n treats its Program Change as a drum-kit selection instead of a melodic instrument. Part 10 (channel 10) defaults to this.

Address	Parameter	Effect
40 1n any other	anything else in Roland's Patch-parameter address space for Part n (Reverb/Chorus Macro, Part Level, Pan, Pitch Key Shift, Velocity Sense, TVF/TVA modifiers, controller-destination depths, ...)	Not individually implemented. Every one of these addresses is instead treated as a write into Part `n's 12-note Scale Tuning table (below), regardless of which parameter the address nominally names. A sequence intended to set, e.g., Reverb Macro or Part Level on real GS hardware will instead corrupt that Part's tuning.
40 1n 40-4B	Scale Tuning, one signed cents offset per semitone class (C...B), applied to Part n	Recognized as such — it's the same fallthrough as the row above, landing on its real address.

Not recognized:

- Reset to GS Standard Mode
- System Parameters (Master Tune, Master Volume-via-**DT1**, Master Key-Shift, Master Pan — address family **40 00 00-06**)
- Display Data (address family **10 00 xx**, Model ID **45**)
- Drum Setup Parameters (address family **41 xx xx** — per-drum-instrument key/level/pan/reverb/chorus edits)
- Bulk Dump (address families **48 xx xx/49 xx xx**)

Any **DT1** (or **RQ1**) message with an address outside the table above, or with a manufacturer ID other than **41/7E/7F**, is accepted and silently dropped.

2. MIDI Implementation Chart

Basic Channel

The MIDI channel(s) each of the 16 Parts answers to.

Mode

Refers to the MIDI Mode messages (Omni On/Off, Mono/Poly). **slopps** always behaves as Omni Off; only the Mono/Poly half of Mode is switchable, per channel.

Note Number / True Voice

The range of note numbers accepted, and the range actually reproduced. Note number 60 is middle C.

Velocity

Whether Note On and Note Off messages' velocity byte affects anything.

After Touch

Keys = polyphonic (per-note) aftertouch, **Ch**s = channel (monophonic) aftertouch.

Control Change

See [Control Change](#) for the full per-controller list — this row just points there.

Program Change

The program numbers a Program Change message can select

(raw message value — one less than the corresponding instrument number as usually printed on an instrument list).

System Exclusive

See [System Exclusive Message](#) for the full address table.

System Common, System Real Time

Sequencer/drum-machine sync messages. Not used, except Reset (FF).

2.1. Implementation Chart

Func	Transmit	Accept	Remarks
Basic Channel Default	×	1 – 16	Part <i>n</i> listens on channel <i>n</i> by default.
Basic Channel Changed	×	1 – 16, each	Via GS RCV CHANNEL.
Mode Default	×	Mode 3	Omni Off, Poly.
Mode Messages	×	Mono, Poly (CC126/127)	Omni On/Off received, no effect.
Mode Altered	×	Mode 3, 4 (m=1)	
Note Number	×	0 – 127	Depends on the loaded gm.dls .
True Voice	×	0 – 127	Depends on the loaded gm.dls .
Pitch Bender	×	O	14-bit, continuous.
Control Change	×	see below	Its own table, directly below this one.
Prog Change	×	O	
True #	×	0 – 127	
System Exclusive	×	O	• GM On/Off • GS Reset • RCV CHANNEL • USE RHYTHM PART • Master Volume • Scale Tuning • Device ID and checksum unchecked
System Common	×	×	
System Real Time	×	×	Except Reset — see Aux Messages, below.
Aux Messages			
Local ON/OFF	×	×	
All Notes OFF	×	O (CC123)	
Active Sense	×	×	
Reset	×	O (FF)	
Velocity			
Note ON	×	O	
Note OFF	×	×	
After Touch			
Key's	×	×	
Ch's	×	×	

2.2. Control Change

CC	Transmit	Accept	Remarks
Bank Select 0 / 32	×	cond.	No effect until a GS Reset has been received.
Mod Wheel 1	×	O	
Data Entry 6 / 38	×	O	Only meaningful with RPN, below.
Channel Volume 7	×	O	Default 100.
Pan 10	×	O	Default 64 (center).
Expression 11	×	O	Default 127.
Hold 1 64	×	O	
Reverb Send 91	×	×	No reverb processor exists.
Chorus Send 93	×	×	No chorus processor exists.
NRPN 98 / 99	×	cond.	Only deselects the current RPN; no NRPN parameter is ever addressable.
RPN 100 / 101	×	O	Pitch Bend Sensitivity / Fine Tune / Coarse Tune only.
All Sound Off 120	×	O	Bypasses a held sustain pedal.
Reset All Ctrl 121	×	O	Bank/Program and RPN selection untouched.
Local Control 122	×	×	No local keyboard to disconnect.
All Notes Off 123	×	O	Honors a held sustain pedal.
Omni Off 124	×	×	
Omni On 125	×	×	
Mono On 126	×	O	Also silences the channel (implicit all-notes-off).
Poly On 127	×	O	Also silences the channel (implicit all-notes-off).
Other —	×	×	No effect, no error.

NOTE

- **O** : recognized/handled.
- **×** : received (or n/a) with no effect.
- **cond.** : recognized, but only under a stated condition — see Remarks.

Mode 1: Omni On, Poly.
Mode 2: Omni On, Mono.
Mode 3: Omni Off, Poly.
Mode 4: Omni Off, Mono.

3. Specifications

Number of Parts

16.

Maximum Polyphony

54 voices (48 sustained under continuous overload)

An automatic accelerated-release policy engages once demand exceeds the pool, freeing voices faster than a plain steal-oldest policy would).

Effects

None.

Wavetable source

A DLS-1 sample collection. (Depicted here is the standard gm.dls.)

Output format

22050 Hz, stereo, 16-bit signed PCM.

4. Presets

NOTE

All tables in this section depicts Microsoft’s gm.dls, slopgs does not provide one. Your chosen DLS file will vary.

4.1. Instruments

Program	Name (bank 0)	Variants (bank MSB: name)
Piano		
0	Piano 1	8: Piano 1 16: Piano 1d
1	Piano 2	8: Piano 2
2	Piano 3	8: Piano 3
3	Honky-tonk	8: Honky-tonk
4	E.Piano 1	8: Detuned EP 1 16: E.Piano 1v 24: 60’s E.Piano
5	E.Piano 2	8: Detuned EP 2 16: E.Piano 2v
6	Harpsichord	8: Coupled Hps. 16: Harpsichord 24: Harpsi.o
7	Clav.	—
Chromatic Percussion		
8	Celesta	—
9	Glockenspiel	—
10	Music Box	—
11	Vibraphone	8: Vibraphone
12	Marimba	8: Marimba
13	Xylophone	—
14	Tubular-bell	8: Church Bell 9: Carillon
15	Santur	—
Organ		

Program	Name (bank 0)	Variants (bank MSB: name)
16	Organ 1	8: Detuned Or.1 16: 60’s Organ 1 32: Organ 4
17	Organ 2	8: Detuned Or.2 32: Organ 5
18	Organ 3	—
19	Church Org.1	8: Church Org.2 16: Church Org.3
20	Reed Organ	—
21	Accordion Fr	8: Accordion It
22	Harmonica	—
23	Bandoneon	—
Guitar		
24	Nylon-str.Gt	8: Ukulele 16: Nylon Gt.o 32: Nylon Gt.2
25	Steel-str.Gt	8: 12-str.Gt 16: Mandolin
26	Jazz Gt.	8: Hawaiian Gt.
27	Clean Gt.	8: Chorus Gt.
28	Muted Gt.	8: Funk Gt. 16: Funk Gt.2
29	Overdrive Gt	—
30	DistortionGt	8: Feedback Gt.
31	Gt.Harmonics	8: Gt. Feedback
Bass		
32	Acoustic Bs.	—
33	Fingered Bs.	—
34	Picked Bs.	—
35	Fretless Bs.	—
36	Slap Bass 1	—
37	Slap Bass 2	—
38	Synth Bass 1	1: SynthBass101 8: Synth Bass 3
39	Synth Bass 2	8: Synth Bass 4 16: Rubber Bass
Strings		
40	Violin	8: Slow Violin
41	Viola	—
42	Cello	—
43	Contrabass	—
44	Tremolo Str	—
45	PizzicatoStr	—
46	Harp	—
47	Timpani	—
Ensemble		
48	Strings	8: Orchestra
49	Slow Strings	—
50	Syn.Strings1	8: Syn.Strings3
51	Syn.Strings2	—
52	Choir Aahs	32: Choir Aahs 2
53	Voice Oohs	—

Progr am	Name (bank 0)	Variants (bank MSB: name)
54	SynVox	—
55	OrchestraHit	—
Brass		
56	Trumpet	—
57	Trombone	1: Trombone 2
58	Tuba	—
59	MutedTrumpet	—
60	French Horns	1: Fr.Horn 2
61	Brass 1	8: Brass 2
62	Synth Brass1	8: Synth Brass3 16: AnalogBrass1
63	Synth Brass2	8: Synth Brass4 16: AnalogBrass2
Reed		
64	Soprano Sax	—
65	Alto Sax	—
66	Tenor Sax	—
67	Baritone Sax	—
68	Oboe	—
69	English Horn	—
70	Bassoon	—
71	Clarinet	—
Pipe		
72	Piccolo	—
73	Flute	—
74	Recorder	—
75	Pan Flute	—
76	Bottle Blow	—
77	Shakuhachi	—
78	Whistle	—
79	Ocarina	—
Synth Lead		
80	Square Wave	1: Square 8: Sine Wave
81	Saw Wave	1: Saw 8: Doctor Solo
82	Syn.Calliope	—
83	Chiffer Lead	—
84	Charang	—
85	Solo Vox	—
86	5th Saw Wave	—
87	Bass & Lead	—
Synth Pad		
88	Fantasia	—
89	Warm Pad	—
90	Polysynth	—
91	Space Voice	—
92	Bowed Glass	—
93	Metal Pad	—
94	Halo Pad	—

Progr am	Name (bank 0)	Variants (bank MSB: name)
95	Sweep Pad	—
Synth Effects		
96	Ice Rain	—
97	Soundtrack	—
98	Crystal	1: Syn Mallet
99	Atmosphere	—
100	Brightness	—
101	Goblin	—
102	Echo Drops	1: Echo Bell 2: Echo Pan
103	Star Theme	—
Ethnic		
104	Sitar	1: Sitar 2
105	Banjo	—
106	Shamisen	—
107	Koto	8: Taisho Koto
108	Kalimba	—
109	Bagpipe	—
110	Fiddle	—
111	Shanai	—
Percussive		
112	Tinkle Bell	—
113	Agogo	—
114	Steel Drums	—
115	Woodblock	8: Castanets
116	Taiko	8: Concert BD
117	Melo. Tom 1	8: Melo. Tom 2
118	Synth Drum	8: 808 Tom 9: Elec Perc.
119	Reverse Cym.	—
Sound Effects		
120	Gt.FretNoise	1: Gt.Cut Noise 2: String Slap
121	Breath Noise	1: Fl.Key Click
122	Seashore	1: Rain 2: Thunder 3: Wind 4: Stream 5: Bubble
123	Bird	1: Dog 2: Horse-Gallop 3: Bird 2
124	Telephone 1	1: Telephone 2 2: DoorCreaking 3: Door 4: Scratch 5: Wind Chimes

Program	Name (bank 0)	Variants (bank MSB: name)
125	Helicopter	1: Car-Engine 2: Car-Stop 3: Car-Pass 4: Car-Crash 5: Siren 6: Train 7: Jetplane 8: Starship 9: Burst Noise
126	Applause	1: Laughing 2: Screaming 3: Punch 4: Heart Beat 5: Footsteps
127	Gun Shot	1: Machine Gun 2: Lasergun 3: Explosion

Note	Sound	Note	Sound
49	Crash Cymbal 1	73	Short Guiro
50	High Tom	74	Long Guiro
51	Ride Cymbal 1	75	Claves
52	Chinese Cymbal	76	Hi Wood Block
53	Ride Bell	77	Low Wood Block
54	Tambourine	78	Mute Cuica
55	Splash Cymbal	79	Open Cuica
56	Cowbell	80	Mute Triangle
57	Crash Cymbal 2	81	Open Triangle
58	Vibraslap		

4.2. Drums

NOTE

For a Part in rhythm mode (see USE RHYTHM PART, [System Exclusive Message](#)), note numbers select individual drum/percussion sounds instead of a pitched instrument — but which sound each note number produces differs per drum kit ([Drum kits](#)).

Available kits:

Program	Kit
0	Standard
8	Room
16	Power
24	Electronic
25	TR-808
32	Jazz
40	Brush
48	Orchestra
56	SFX

Standard kit maps to:

Note	Sound	Note	Sound
35	Acoustic Bass Drum	59	Ride Cymbal 2
36	Bass Drum 1	60	Hi Bongo
37	Side Stick	61	Low Bongo
38	Acoustic Snare	62	Mute Hi Conga
39	Hand Clap	63	Open Hi Conga
40	Electric Snare	64	Low Conga
41	Low Floor Tom	65	High Timbale
42	Closed Hi-Hat	66	Low Timbale
43	High Floor Tom	67	High Agogo
44	Pedal Hi-Hat	68	Low Agogo
45	Low Tom	69	Cabasa
46	Open Hi-Hat	70	Maracas
47	Low-Mid Tom	71	Short Whistle
48	Hi-Mid Tom	72	Long Whistle